



NOVA HOME

DECI COMPETITION LEVEL 3 — UI/UX SPECIFICATION

NOVA Home UI/UX Design System & Architecture

Comprehensive UX Guidelines, Color Tokens, Typography Scale, Component Specs, and Wireframe Layouts

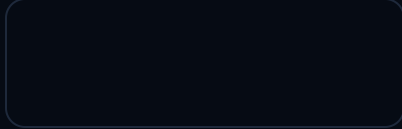
1. Executive Summary & UX Vision

NOVA Home is designed as an ultra-modern, high-impact intelligent smart home engineering platform. The UI/UX vision centers on **Instant Cognitive Clarity**, **High-Tech Dark Aesthetics**, and **Explainable AI Interaction**.

Evaluators and homeowners can view critical telemetry (Temperature, Humidity, Ambient Light, Motion) and perimeter security status within 3 seconds of opening the interface, with instantaneous low-latency control over IoT actuators.

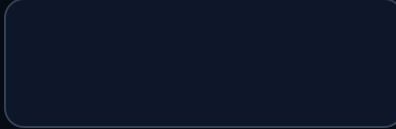
2. Color Palette & Design Tokens

A curated dark navy background theme with electric cyan glowing accents, glassmorphic container overlays, and clear functional semantic colors.



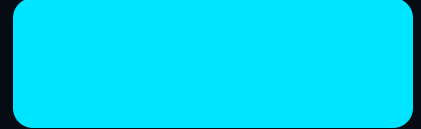
Deep Navy BG

#060B14



Glass Card Surface

#0E172A (85%)



Electric Cyan Accent

#00E5FF



Secure / Optimal

#10B981 / #34D399



Perimeter Alert

#EF4444 / #F87171



AI Action Warning

#F59E0B / #FBBF24



Autonomous AI

#A855F7 / #C084FC

3. Typography System

Role / Target	Font Family	Size / Weight	Usage Example
Brand / Titles	Plus Jakarta Sans	2.5rem (40px) / 800 Bold	Main Section Headers & Logo Brand Name
Subtitles & Titles	Plus Jakarta Sans	1.2rem (19px) / 700 SemiBold	Card Titles & Section Descriptions
Sensor Values	Fira Code	2.2rem (35px) / 800 Bold	Real-Time Sensor Values (28.5°C, 54%)
Body Text	Inter	0.95rem (15px) / 400 Regular	General Paragraphs & AI Condition Explanations
Pin Map & Terminal	Fira Code	0.85rem (13px) / 700 Bold	GPIO Pin Badges & Live MQTT Terminal Logs

4. UI Component Architecture

SECURITY PANEL COMPONENT

Dynamic banner monitoring physical perimeter states. Toggles between **SYSTEM SECURE** (Emerald Green) and **PERIMETER ALERT** (Pulsing Red) when doors/windows are opened.

SENSOR TELEMETRY CARD

Includes FontAwesome category icon, sensor badge (e.g. DHT22), numeric value in Fira Code font, horizontal progress gauge, and thermal/moisture status message.

ACTUATOR CONTROL ITEM

Integrates custom rounded toggle switches, PWM brightness/speed sliders, state labels, and active visual glowing feedback upon activation.

EXPLAINABLE AI MODULE

Provides transparent AI reasoning via Explain Condition, Suggested Action, rule tags, and a 1-click Apply AI Action button.

5. Accessibility & Information Architecture

Navigation Structure: Sticky top header bar with active scroll-spy detection highlighting current viewports (Dashboard, Architecture, Hardware Flow, AI Pipeline, GPIO Map).

Accessibility Compliance: Exceeds WCAG 2.1 AA standard for color contrast ratio (> 7:1 for main text). High-contrast icons accompany color badges to ensure color-blind accessibility.